Hands-On Session 1 – 2 - 3

Machine & Deep Learning Applications in Aquatic Organisms' Multi-Omic Data

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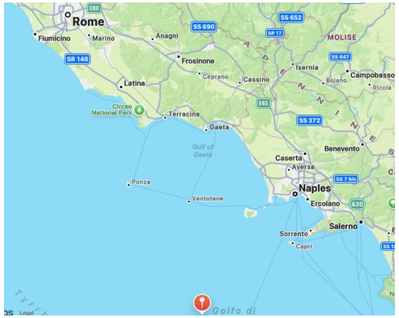
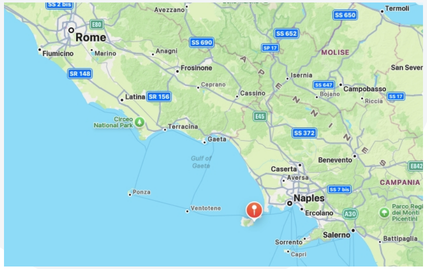
Contents

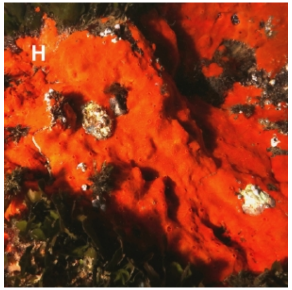
Chosen Data

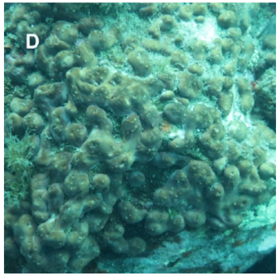
Species	Number of Samples	Site
C. crambe	5-2	GF-SA
C. nucula	5-5	GF-SA
C. reniformis	5-5	GF-SA
P. ficiformis	3-5	GF-SA

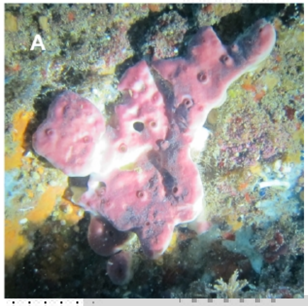
GF: Grotta Mago SA: Sant'Anna











Session 1: ML in Microbiome and Multiomic Data

General Look at AI in Microbiome

- ML-DL workflow:
  - Biological question
  - Setting – classification, regression, clustering…
  - Integration level

- QC:
  - ROC-AUC – test & training sets
  - PR-AUC
  - Accuracy – Error rate

- Obtain:
  - Filter down possibilities
  - Potential biomarkers
  - Effect sizes

AI in Microbiome Studies

Disease detection	CNN/autoencoder	Oh, M., & Zhang, L. (2020)
Large scale annotation	large-sized omic integration with ML annotation	Carradec, Q. et al (2018)
Microbial community and temporal dynamics prediction	wastewater treatment plants	Andersen, K. S. *et al* (2025)
AI based feeding technique - Shrimp biomass detection	Image-based deep learning analysis	Darapaneni, N. *et al* (2022)
Image-based disease detection	SVM (Support Vector Machines)	Chen F. *et al* (2022)
Water quality prediction	artificial neural networks	Chen, Y. *et al* (2020)
Fish species classification	Multi-class SVM	Hu et al (2012)

Session 1: Hands-On

Session 2: DL in Microbiome and Multiomic Data

Deep Learning Approaches

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Neural Networks (Recurrent, Graph and Convolutional)

Long Short-Term Memory Network (LSTM)

Variational Auto Encoders

Neural Matrix Factorization

SHAP/Integrated Gradients*

DIABLO (MixOmics)**

Similarity Network Fusion Method**

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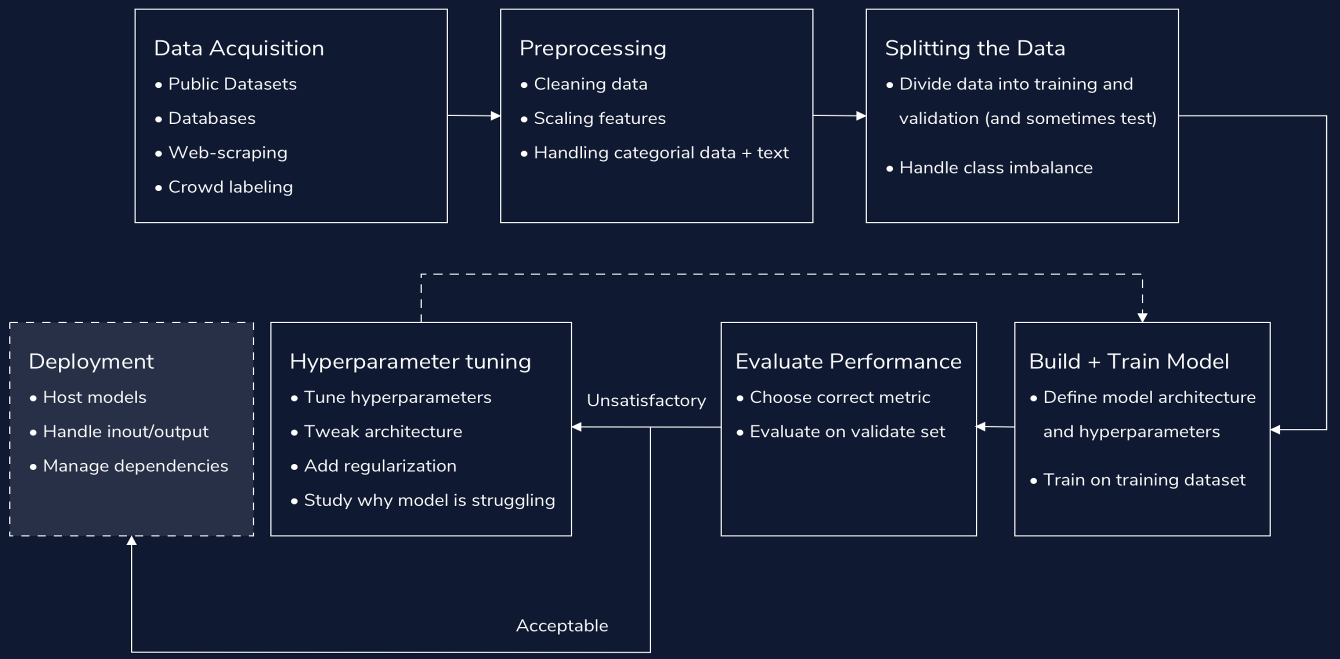
Session 2: Deep Learning Workflows, Microbiome & Multiomics



- Sparse data
- Few samples per study
- Biological noise
- High dimensionality
- Different scales noise, and feature spaces
- Lack of standardization
- Biological bias and knowledge gaps
- Mislabels
- Data leakage

Session 2: Hands-On

Session 3: Microbiome & Metagenome Analysis Pipeline: Data to ML/DL Workflows

16S rRNA Analysis Workflow

- Data QC
- Trimming, Error correction and Filtering
- Denoising
- ASV/OTU Clustering
- Taxonomic Classification - Annotation
- Alpha-Beta Diversity Calculation

WMS Analysis Workflow

- Data QC
- Trimming, Error Correction
- Host filtering
- MAG Creation
- MAG QC
- MAG Coverage – Abundance Calculation
- MAG Binning
- MAG Bin QC
- Gene Annotation
- Short read Taxonomic Classification
- Functional Annotation
- PCA-PcoA-ML-DL Workflows

Session 3: Hands-On

Thank you for listening…

Any questions?

References

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